Round 7: GenAI Task-to-Model Comparison and Multimodal Proof of Concept

MoinSystems AI | AI Engineer Internship | Round 7: Generative AI

1. Purpose

This document maps common MoinSystems business tasks to suitable GenAI models, APIs or modalities, and describes one small multimodal proof of concept (PoC). Every result in the "What we observed" column comes from a test run during this round. Sample sizes are small (see Section 5), so treat results as indicative, not final.

2. Task-to-Model Comparison

Business task	Modality	Suggested model / API	Why	What we observed
Classify a customer message and extract name, budget, urgency	Text, structured output	gpt-4o-mini for simple extraction; gpt-4o when date reasoning is needed	A Pydantic schema forces valid JSON, so output can go straight into n8n or a CRM	On 4 test emails, gpt-4o-mini got 3 of 4 fully right (it misjudged urgency for "end of the month"); gpt-4o got 4 of 4
Answer customer questions about the company	Text + retrieval	LLM + RAG (Qdrant, Round 2)	A plain LLM does not know private company data and may hallucinate	Covered in Rounds 1 and 2
Turn a customer voice note into a lead	Audio to text to structured output	Whisper (whisper-1) + gpt-4o	Whisper transcribes; the LLM extracts fields	Name, budget, deadline and phone number (with leading 0) were all correct on an English recording
Speak a reply to the customer	Text-to-speech	gpt-4o-mini-tts	Controllable voice and tone through an instructions field	Voice sounded almost real. Changing the voice and tone settings was not tested
Extract a lead from a screenshot or business card	Image input (vision)	gpt-4o with an image in the message	The same Lead schema works for image input	All six fields correct on a test screenshot
Create marketing posters and banners	Image generation	gpt-image-2 (API); Microsoft Copilot as a free option	Text-to-image and image editing for visuals	Copilot followed the prompts, spelled all text correctly and changed only the background in an edit; gpt-image-2 followed the "no text" and layout instructions in a detailed prompt
Short promotional video	Video generation	Tools such as Veo, Runway, Kling, Luma (mostly paid or limited)	Useful for occasional marketing content	Not tested hands-on. Per OpenAI's own page, the Sora product has not been available since April 26, 2026
Calculate invoice totals	None (not an LLM job)	Plain Python code	LLMs are unreliable with large arithmetic	Not tested; known limitation

3. Multimodal Proof of Concept

What it does: accepts either a customer voice recording or a screenshot of an inquiry, extracts a structured lead, builds a reply, and speaks the reply as an audio file.

Flow:

1. Input file: audio (.m4a, .mp3, .wav) or image (.png).
2. Audio goes to Whisper for a transcript; an image goes straight to gpt-4o.
3. gpt-4o returns a validated Lead (intent, name, phone, budget, currency, urgency).
4. Plain Python builds a reply: an apology for a complaint, a thank-you for anything else.
5. gpt-4o-mini-tts turns the reply into reply.mp3.

Test results:

Input	Extracted lead	Correct?
English voice note (Amna, boutique chatbot)	inquiry, Amna, 0599223310, 10000, AED, medium	Yes. Currency first came back as the word "Dirham"; after an ISO-code rule was added to the prompt, a re-run returned AED
Screenshot (Sara, salon booking chatbot)	inquiry, Sara, 0501234567, 8000, AED, medium	Yes

The PoC uses three input and output types (audio, image, text) and runs on a single OpenAI API key.

4. Model Selection Guide

Factor	Question to ask	Example from this round
Capability	Can the model do this task at all?	Reading a screenshot needs a model with image input
Quality	How accurate does the answer need to be?	gpt-4o-mini made one date-based urgency error that gpt-4o did not
Latency	How fast must the answer arrive?	A live chat needs fast replies; a nightly batch does not
Cost	How many requests per day?	Thousands of emails a day favour the cheaper model; check current rates on OpenAI's pricing page, which changes over time

Rule of thumb: start with the cheaper model, test it on real examples, and move to a larger model only for the tasks where it fails. Do arithmetic and date calculations in code whenever possible.

5. Limitations and Findings

- Small samples: 4 emails, 1 voice recording and 1 screenshot. A real decision needs 20 to 30 varied examples.
- Prompt rules matter as much as the schema: the schema fixed the format, but intent labels and urgency rules had to be written explicitly (a missing "complaint" rule caused wrong labels until it was restored).
- Inconsistent currency format: the same currency came back as "Dirham" in one run and "AED" in another. Fixed: an ISO-code rule was added to the prompt, and a re-run returned AED.
- Not tested: Whisper on Urdu or mixed Urdu-English speech, changing the TTS voice and tone, and API cost and latency measurements. Windows voice typing returned Roman Urdu for Urdu speech.
- Video: concept only. Tools are mostly paid, short-form, and not suited to everyday support tasks.
- Outputs can vary: the same email gave different labels across runs until temperature was set to 0.